
Wireless Communications at the Physical Layer

Part 2: Techniques

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Outline

- Error compensation mechanisms
- Multiplexing at the physical layer
- Modulation
- Spread spectrum

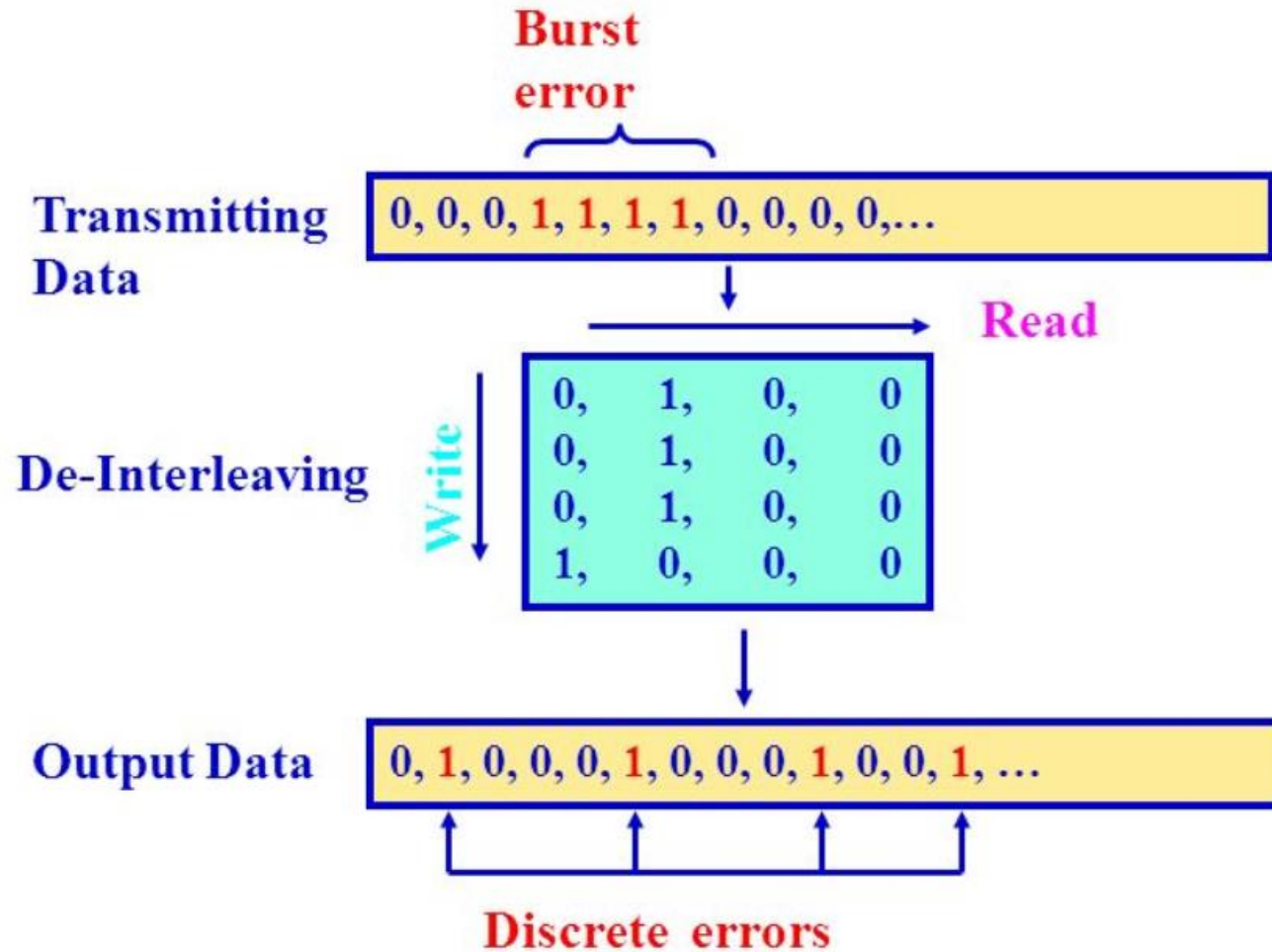
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- Forward error correction
 - Adaptive equalization
 - Diversity techniques

Error compensation mechanisms

Forward Error Correction (FEC)

- Transmitter adds error-correcting code to data block
 - Code is a function of the data bits
- Receiver calculates error-correcting code from incoming data bits
 - If calculated code matches incoming code, no error occurred
 - If error-correcting codes don't match, receiver attempts to determine bits in error and correct them
- k data bits, $n - k$ code bits $\Rightarrow$ frame of n bits
 - FEC k/n
 - Example: FEC 1/3, FEC 2/3

Interleaving



Adaptive equalization

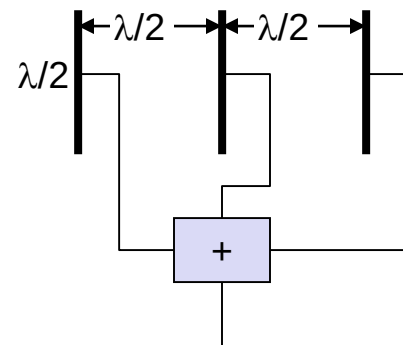
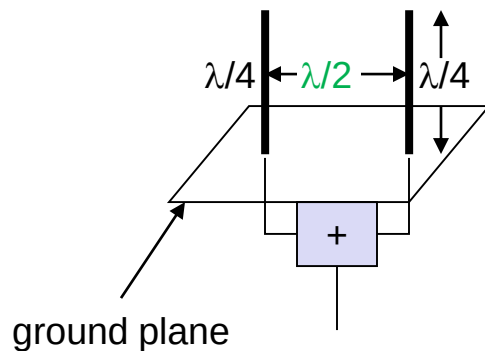
- Can be applied to transmissions that carry analog or digital information
- Used to combat intersymbol interference
- Involves gathering dispersed symbol energy back into its original time interval
- Typically, the transmitter sends a known sequence of bits (training sequence) periodically. The receiver compares the received TS with the expected TS in order to calculate the equalizations coefficients.

Diversity techniques

- Diversity is based on the fact that individual channels experience independent fading events
- Space/Antenna diversity – techniques involving different physical transmission paths
- Frequency diversity – techniques where the signal is spread out over a larger frequency bandwidth or carried on multiple frequency carriers
- Time diversity – techniques aimed at spreading the data out over time (e.g. repetition, interleaving)

Antenna diversity

- Grouping of 2 or more antennas
 - multi-element antenna arrays
- Antenna diversity
 - switched diversity, selection diversity
 - receiver chooses antenna with largest output
 - diversity combining
 - combine output power to produce gain



Multiplexing at the physical layer

Multiplexing

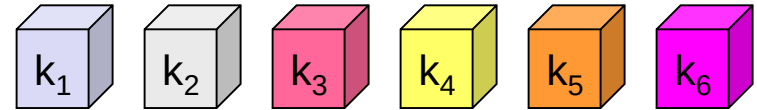
- Multiplexing in 4 dimensions

- space (s_i)
- time (t)
- frequency (f)
- code (c)

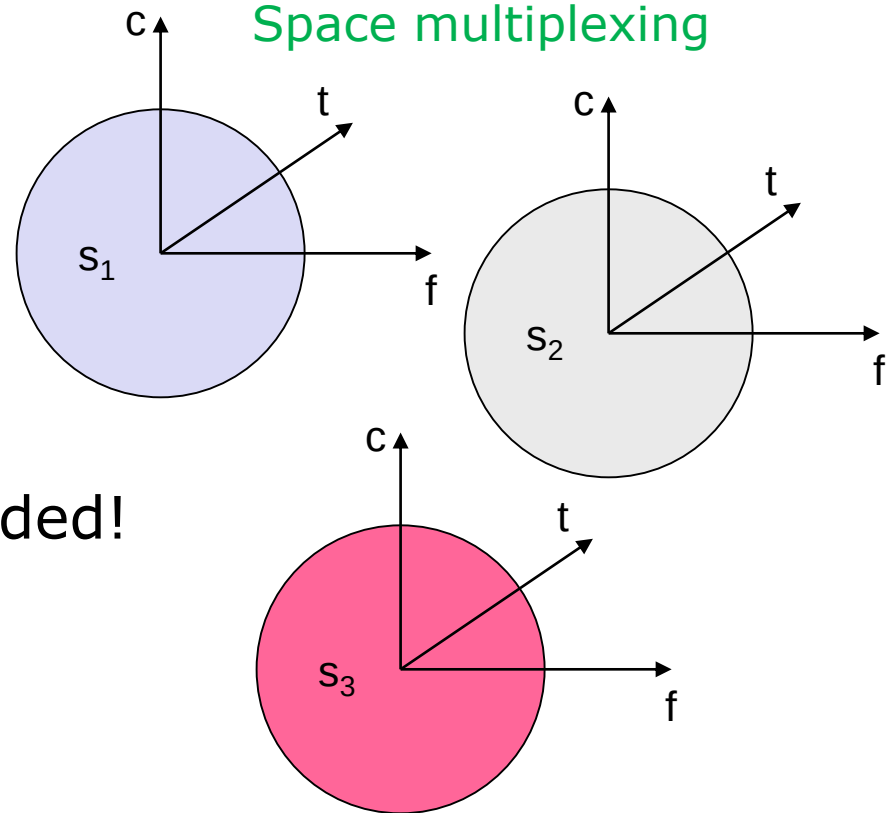
- Goal: multiple use of a shared medium

- Important: guard spaces needed!

channels k_i

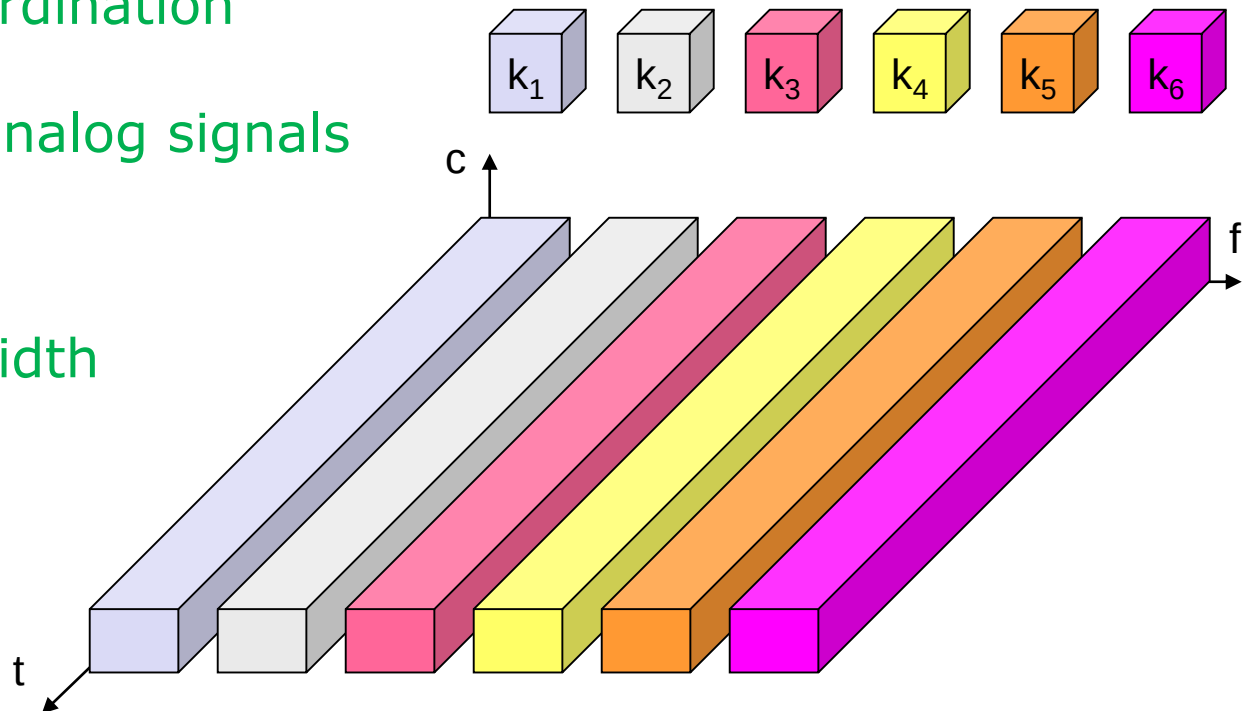


Space multiplexing



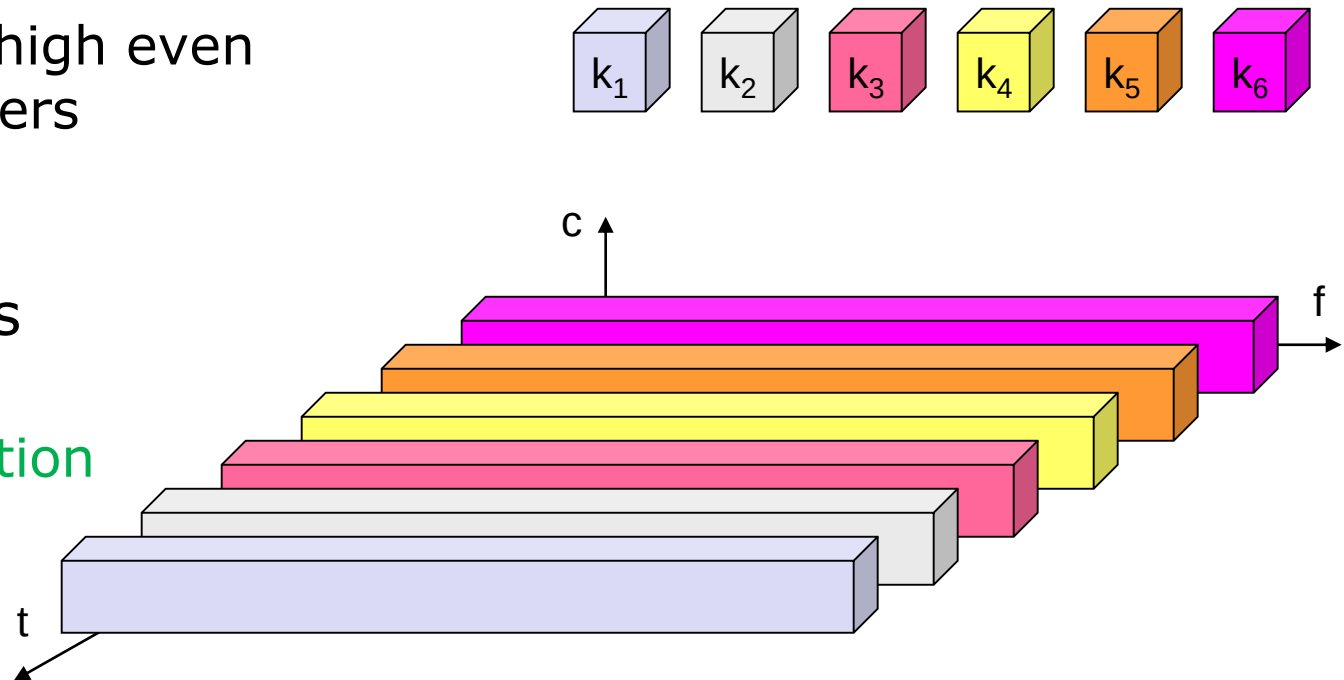
Frequency multiplexing (FDM)

- Separation of the whole spectrum into smaller frequency bands
- A channel gets a certain band of the spectrum for the whole time
- Advantages
 - no dynamic coordination necessary
 - works also for analog signals
- Disadvantages
 - waste of bandwidth if the traffic is distributed unevenly
 - inflexible



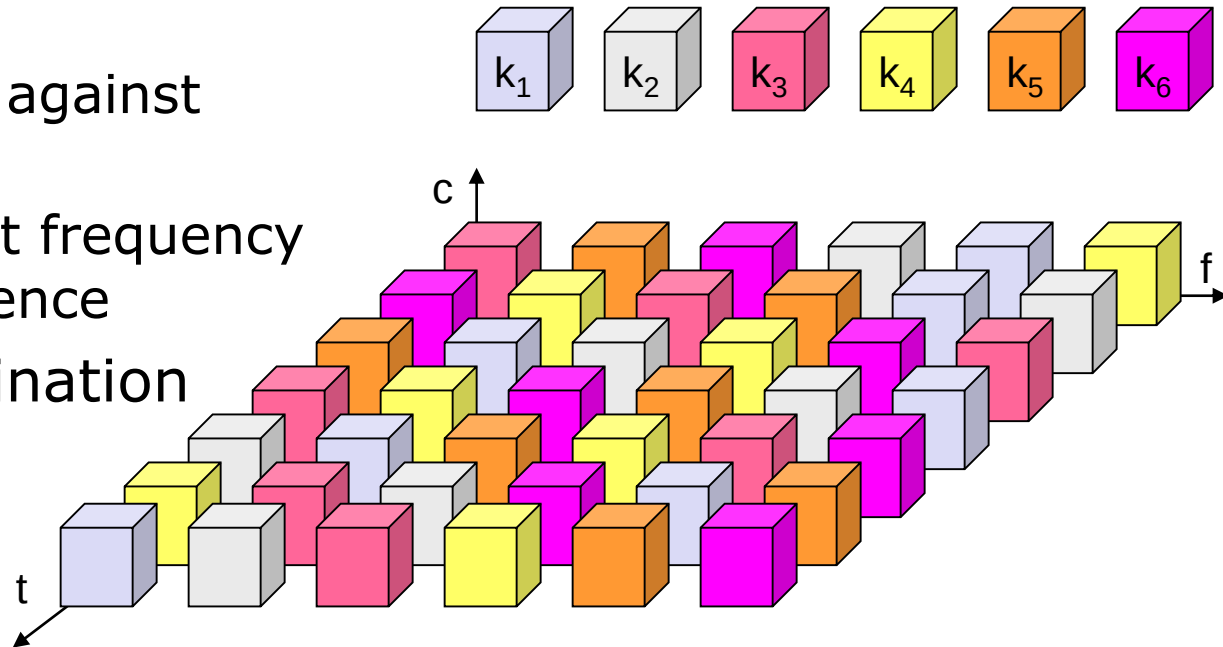
Time multiplexing (TDM)

- A channel gets the whole spectrum for a certain amount of time
- Advantages
 - only one carrier in the medium at any time
 - throughput high even for many users
 - flexible
- Disadvantages
 - precise synchronization necessary



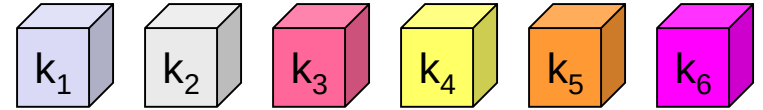
Time and frequency multiplexing

- Combination of both methods
- A channel gets a certain frequency band for a certain amount of time
- Example: GSM
- Advantages
 - better protection against tapping
 - protection against frequency selective interference
- but: precise coordination required



Code multiplexing (CDM)

- Each channel has a unique code



- All channels use the same spectrum at the same time

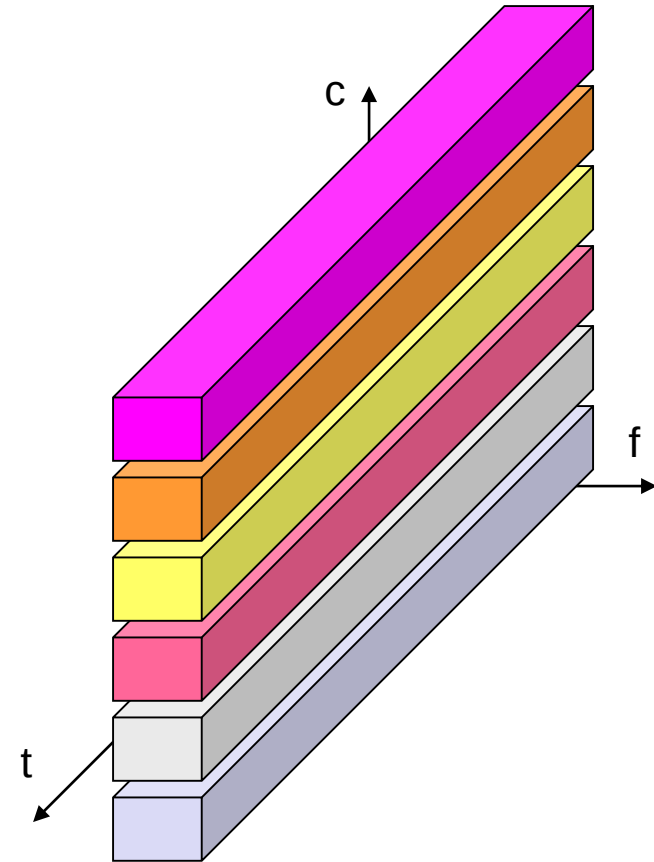
- Advantages

- bandwidth efficient
- no coordination and synchronization necessary
- good protection against interference and tapping

- Disadvantages

- varying user data rates
- more complex signal regeneration

- Implemented using spread spectrum technology



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- Introduction
 - Analog modulation techniques
 - Digital modulation techniques

Modulation

Modulation

- Basics: Transmitter sends a modulated radio wave, receiver can detect whether such a wave is present and also its parameters
- Parameters of a wave = sine function (**carrier**):

$$s(t) = A(t) \sin(2\pi f(t)t + \phi(t))$$

- Parameters: amplitude $A(t)$, frequency $f(t)$, phase $\phi(t)$
- Manipulating these three parameters allows the sender to **express data**; receiver reconstructs data from signal

Receiver: Demodulation

- The receiver looks at the received wave form and matches it with the signal that caused the transmitter to generate this wave form
 - Necessary: one-to-one mapping between data and wave form
- Problems caused by
 - **Carrier synchronization: frequency** can vary between sender and receiver (drift, temperature changes, aging, ...)
 - **Bit synchronization** (actually: symbol synchronization):
When does symbol representing a certain bit start/end?
 - Frame synchronization (is a concern of data link layer):
When does a packet start/end?
 - **Biggest problem**: Received signal is ***not*** the transmitted signal (can suffer deterioration during transmission).

Motivations for modulation

- An antenna must have a size the same order of magnitude of the signal's wavelength to be effective
- Modulation enables frequency division multiplexing
- The propagation characteristics of a medium (e.g. attenuation) vary with the frequency of the signal. Modulation allows the choice of an adequate carrier frequency

Analog Modulation Techniques

- Analog data to analog signal
 - Amplitude modulation (AM)
 - Angle modulation
 - Frequency modulation (FM)
 - Phase modulation (PM)

Amplitude Modulation

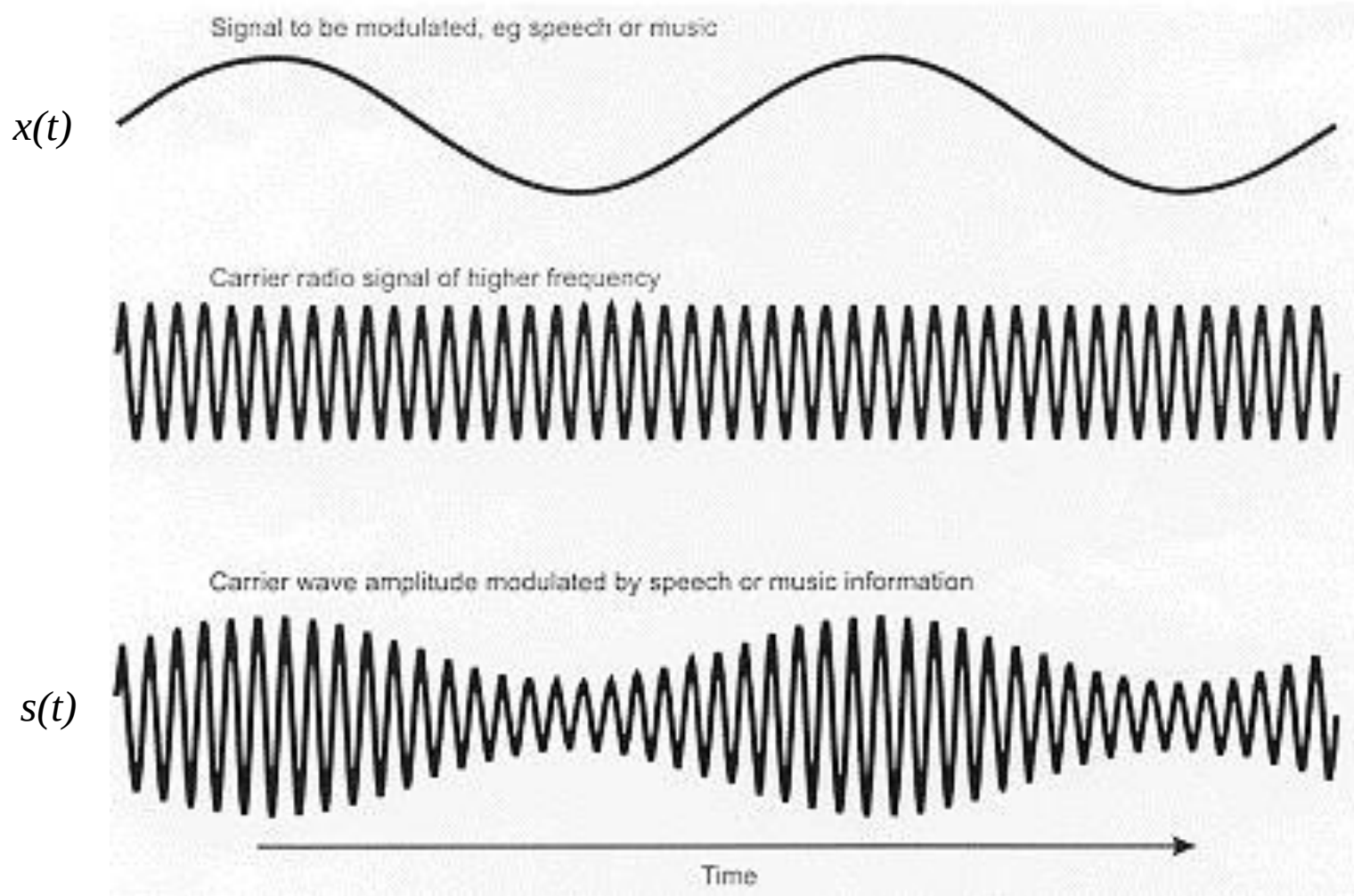
- Amplitude Modulation

$$s(t) = [1 + n_a x(t)] \cos 2\pi f_c t$$

- $c(t) = \cos 2\pi f_c t = \text{carrier}$
- $x(t) = \text{input signal}$
 - $\text{modulating signal} \rightarrow x(t) = m(t)$
- $n_a = \text{modulation index}$
 - Ratio of amplitude of input signal to carrier
- AM modulation principle
 - Uses quadratic (non-linear) function
 - Filters only desired frequency components

$$(m + c)^2 = m^2 + \mathbf{2mc} + c^2$$

Waveform of an AM signal



Amplitude Modulation

- Given a sinusoidal modulating signal

$$x(t) = \cos 2\pi f_m t$$

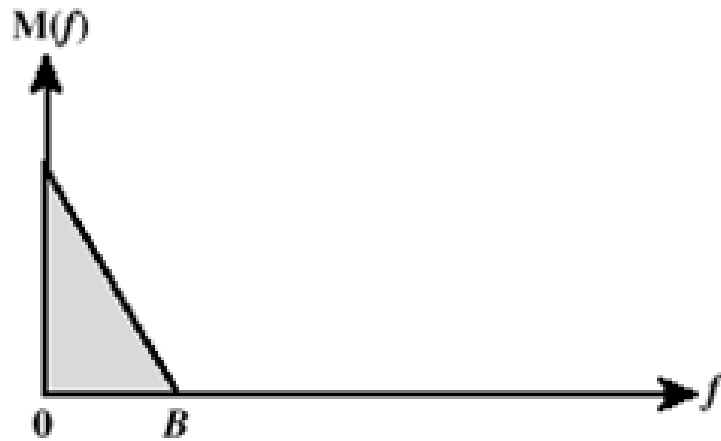
- By using the trigonometric identity

$$\cos(a) \cos(b) = \frac{1}{2} (\cos(a+b) + \cos(a-b))$$

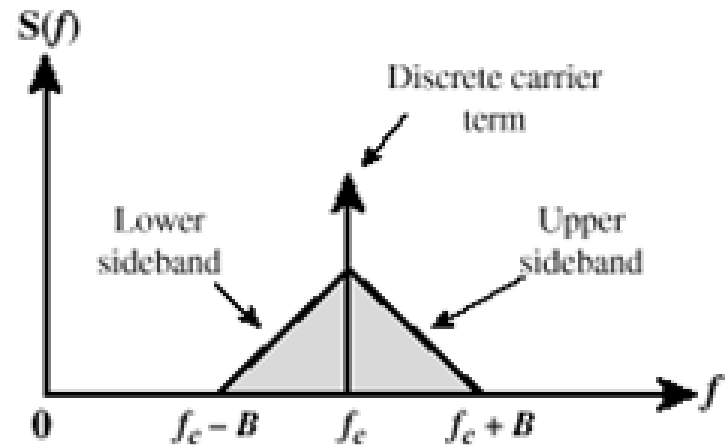
- The modulated signal can be expressed as

$$s(t) = \cos 2\pi f_c t + \frac{n_a}{2} \cos 2\pi (f_c - f_m) t + \frac{n_a}{2} \cos 2\pi (f_c + f_m) t$$

Spectrum of an AM signal



(a) Spectrum of modulating signal

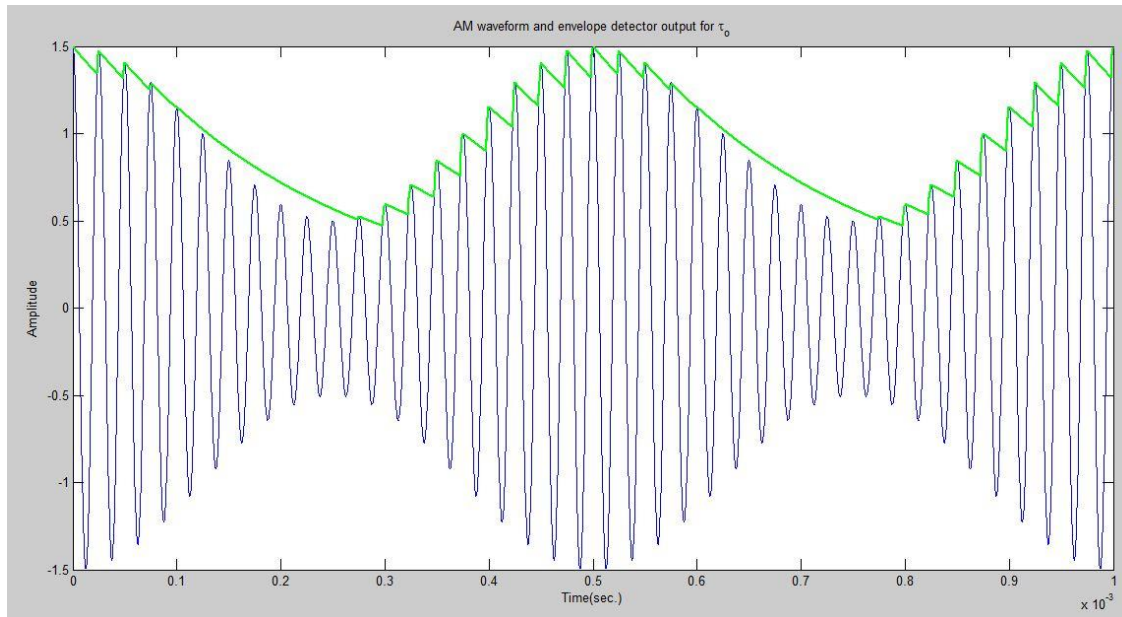
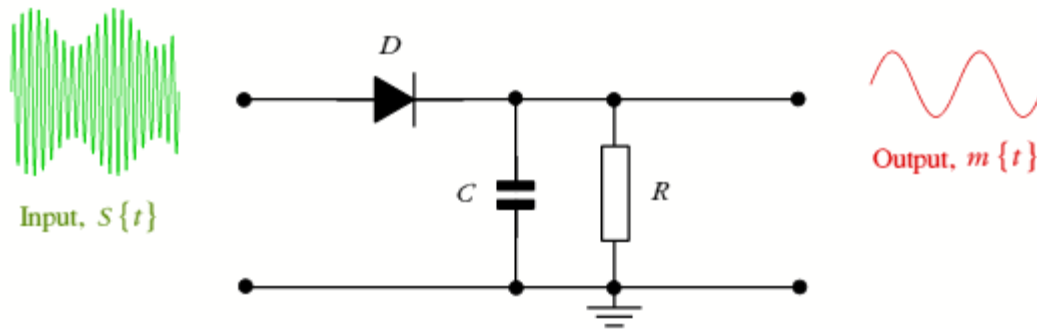


(b) Spectrum of AM signal with carrier at f_c

- Above example is known as AM-DSB – Double Sideband

Envelope detector

- Basic type of AM demodulator



Other AM variants

- Single Sideband (SSB)
 - Sends only one sideband
 - Eliminates other sideband and carrier
 - Advantages
 - Only half the bandwidth is required
 - Less power is required
 - Disadvantages
 - Suppressed carrier can't be used for synchronization purposes
- Double sideband suppressed carrier (DSB-SC): filters out the carrier frequency and sends both sidebands.
- Vestigial sideband (VSB): uses one sideband and reduced-power carrier and a vestige of the other sideband (used by TV broadcasting)

Angle Modulation

$$s(t) = A_c \cos[2\pi f_c t + \phi(t)]$$

- Phase modulation
 - Phase is proportional to modulating signal

$$\phi(t) = n_p m(t)$$

- n_p = phase modulation index
- Frequency modulation
 - Derivative of the phase is proportional to modulating signal

$$\phi'(t) = n_f m(t)$$

- n_f = frequency modulation index

Angle Modulation

- Compared to AM, FM and PM result in a signal whose bandwidth:
 - is also centered at f_c
 - but has a pattern that is much different
 - Angle modulation includes $\cos(\phi(t))$ which produces a wide range of frequencies

Digital modulation techniques

- Modulation using digital signals is known as Shift Keying

- **Amplitude Shift Keying (ASK):**

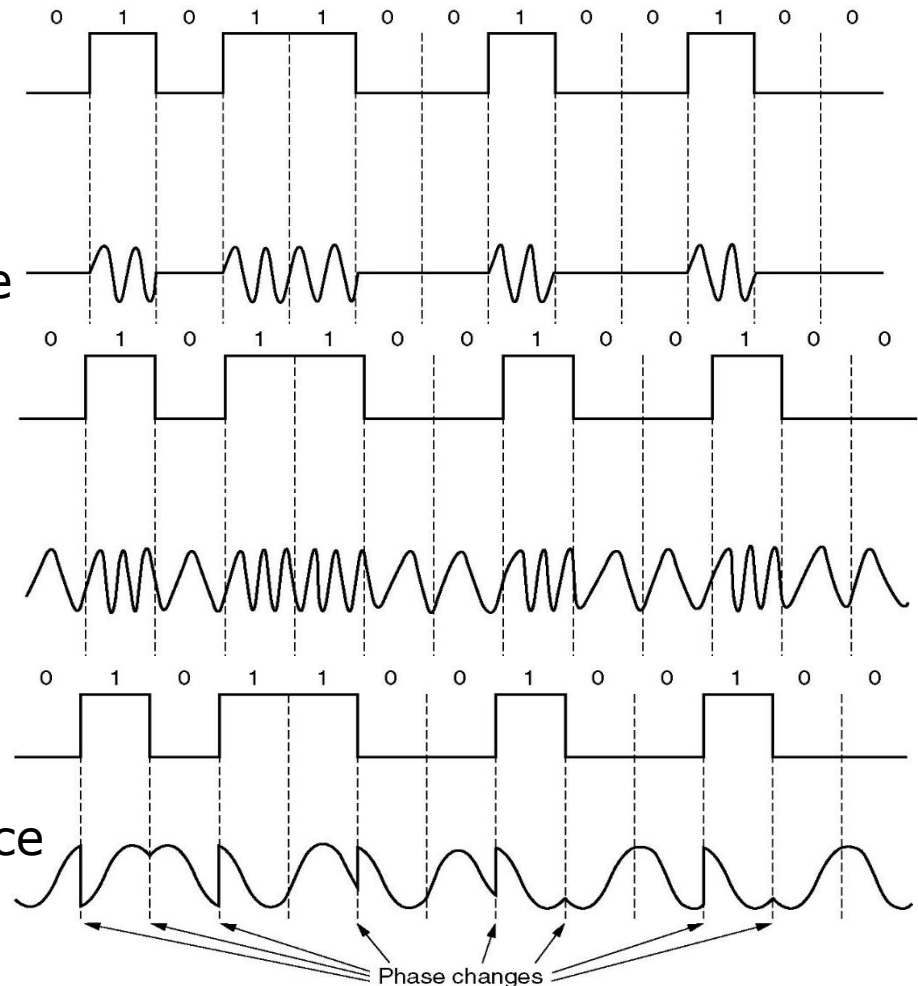
- very simple
- low bandwidth requirements
- very susceptible to interference

- **Frequency Shift Keying (FSK):**

- needs larger bandwidth

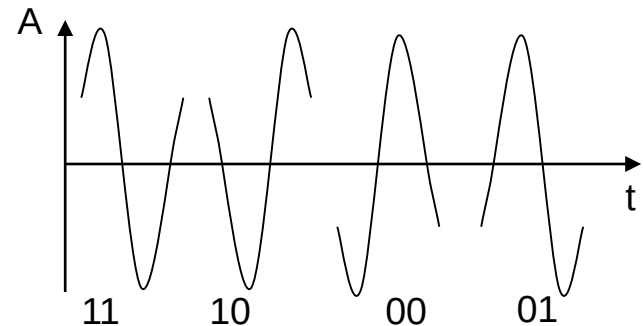
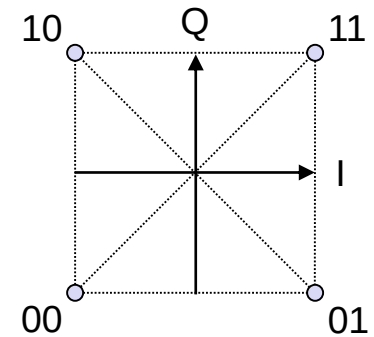
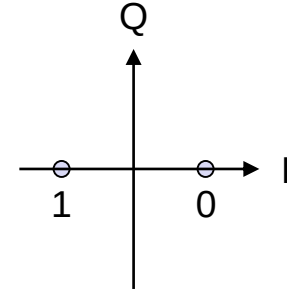
- **Phase Shift Keying (PSK):**

- more complex
- more robust against interference



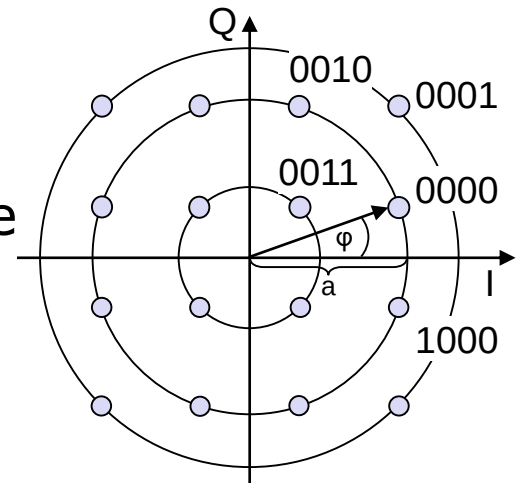
Advanced Phase Shift Keying

- **BPSK (Binary Phase Shift Keying):**
 - bit value 0: sine wave
 - bit value 1: inverted sine wave
 - very simple PSK
 - low spectral efficiency
 - robust, used e.g. in satellite systems
- **QPSK (Quadrature Phase Shift Keying):**
 - 2 bits coded as one symbol
 - symbol determines shift of sine wave
 - needs less bandwidth compared to BPSK
 - more complex
- Often also transmission of relative, not absolute phase shift: DPSK - Differential PSK

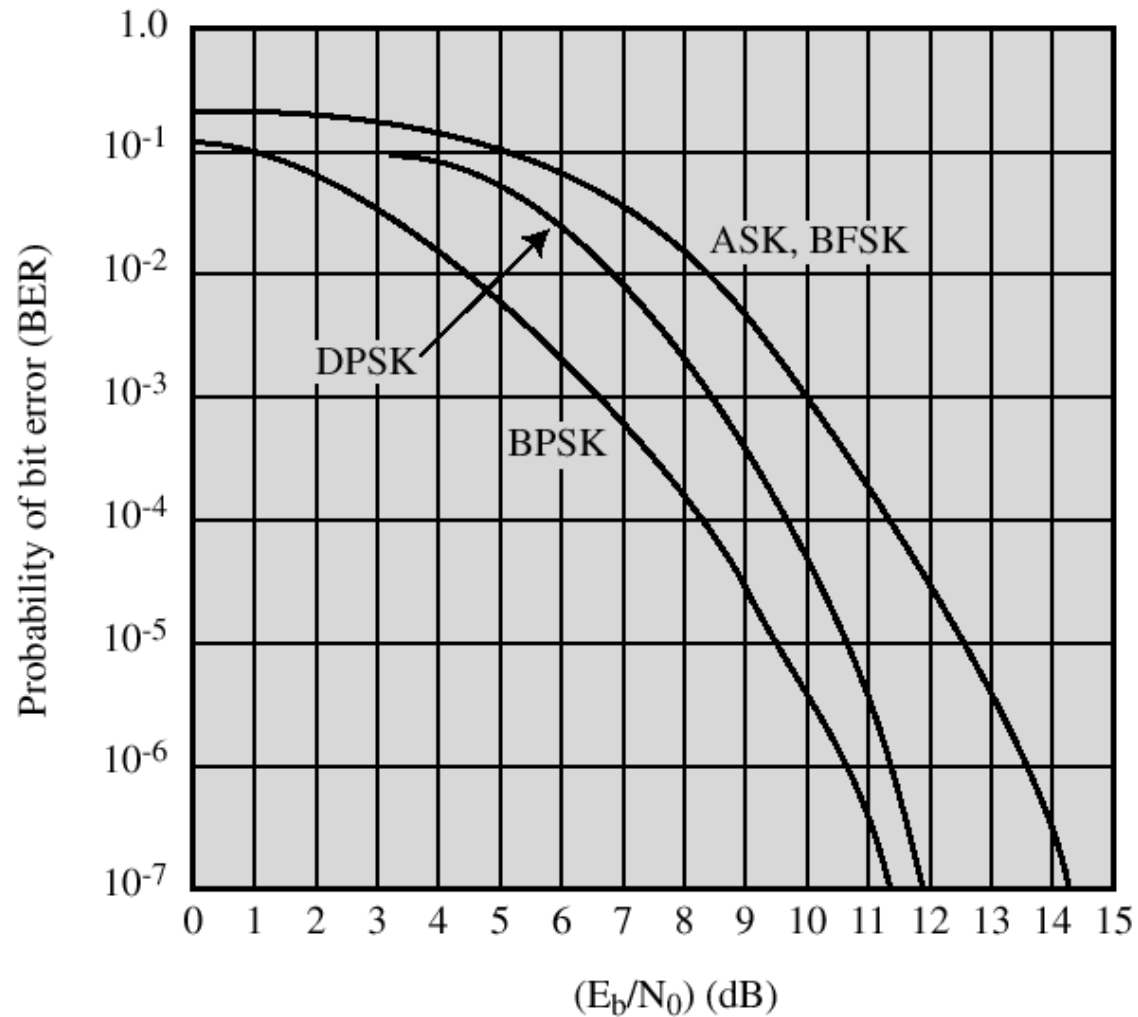


Quadrature Amplitude Modulation

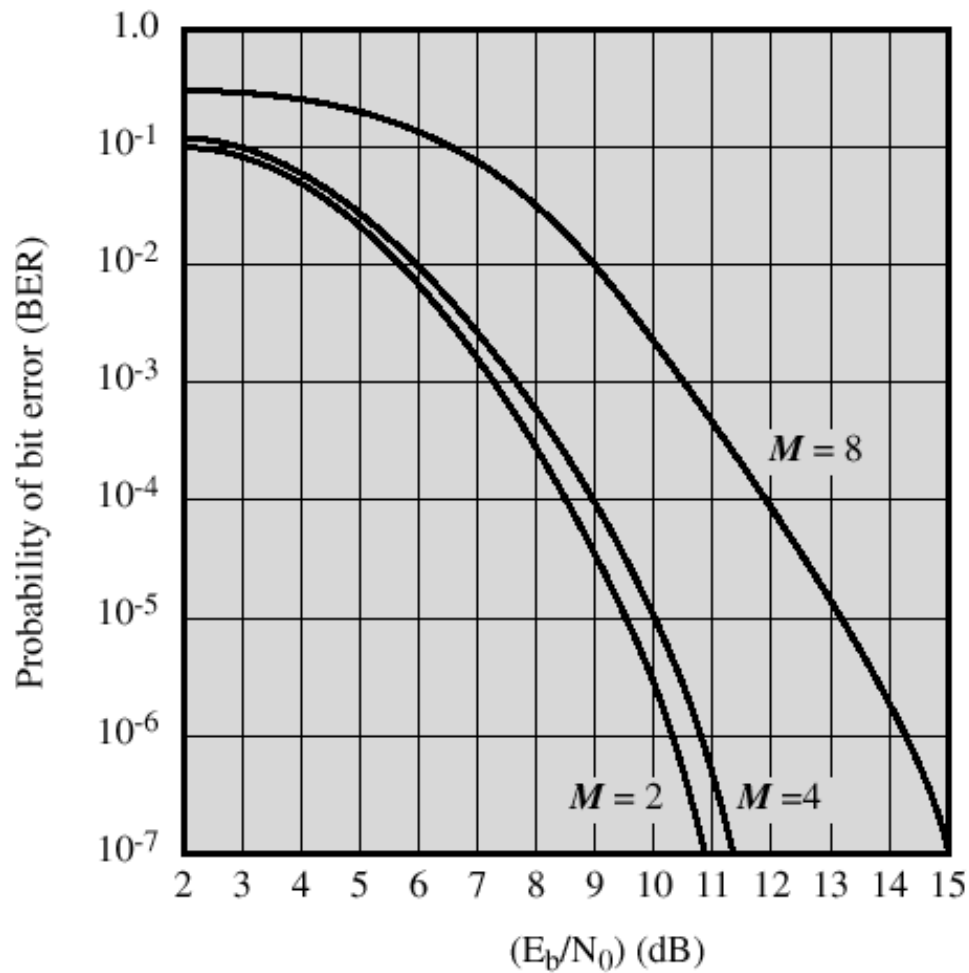
- Quadrature Amplitude Modulation (QAM)
 - combines amplitude and phase modulation
 - it is possible to code n bits using one symbol
 - $M=2^n$ discrete levels, $n=2$ identical to QPSK
- Bit error rate increases with n , but less errors compared to comparable PSK schemes
 - Example: 16-QAM (4 bits = 1 symbol)
 - Symbols 0011 and 0001 have the same phase φ , but different amplitude ; symbols 0000 and 1000 have different phase, but same amplitude.
- Constellation diagram – Representation of modulation symbols on plane



Performance of binary modulations



Performance of multilevel PSK

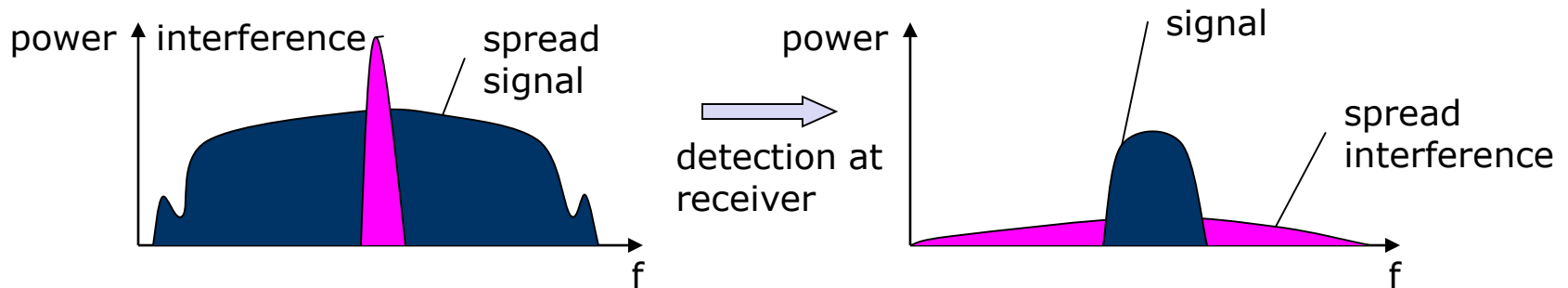


(b) Multilevel PSK (MPSK)

Spread spectrum

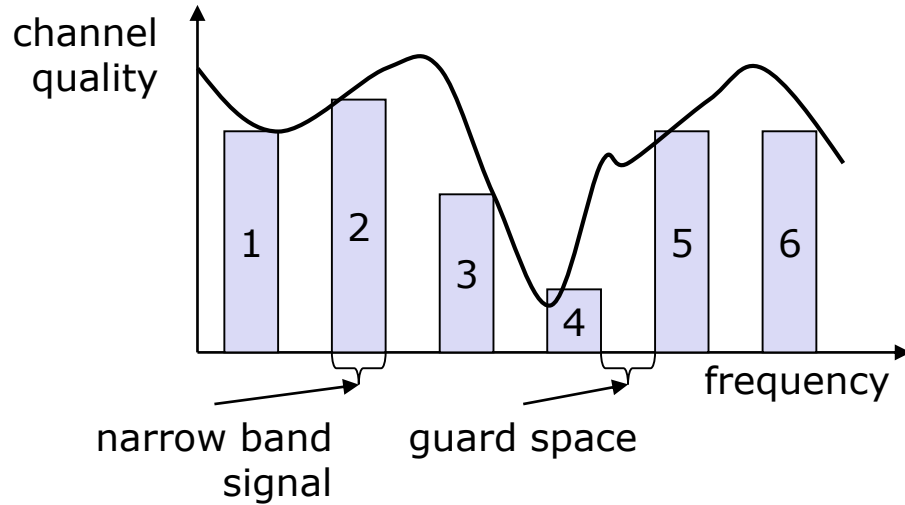
Spread spectrum technology

- Problem of radio transmission: frequency dependent fading and interference can wipe out narrow band signals
- Solution: spread the narrow band signal into a broad band signal using a special code
 - protection against narrow band interference

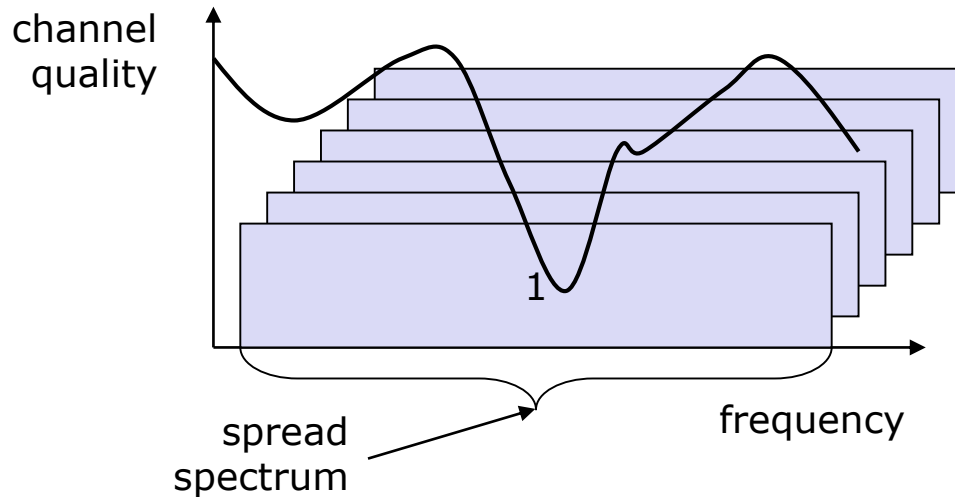


- Side effects:
 - coexistence of several signals without dynamic coordination
 - tap-proof
- Modalities: Direct Sequence, Frequency Hopping

Spreading and frequency selective fading



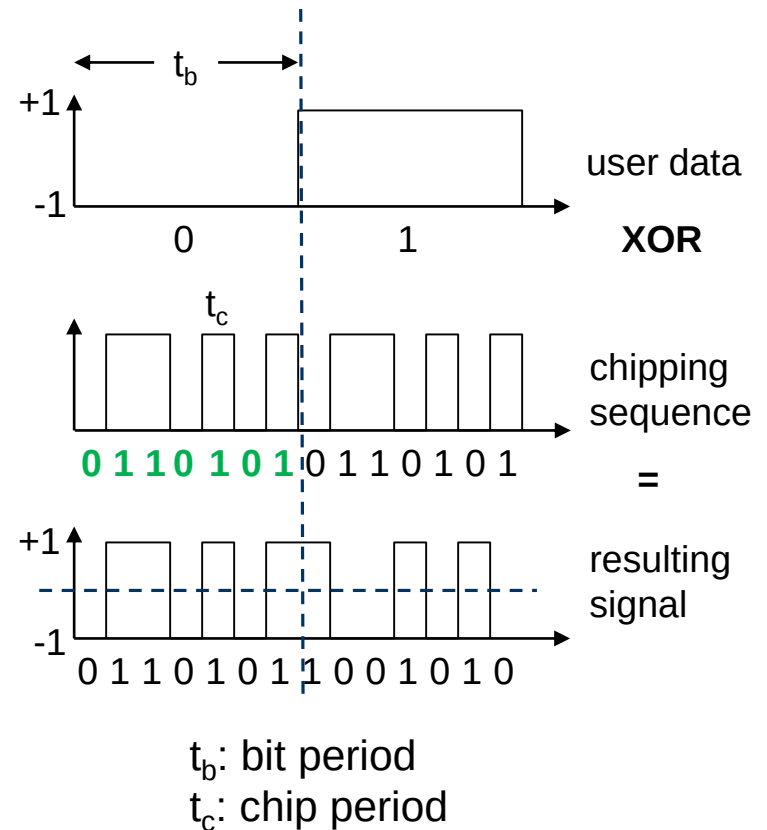
narrowband channels



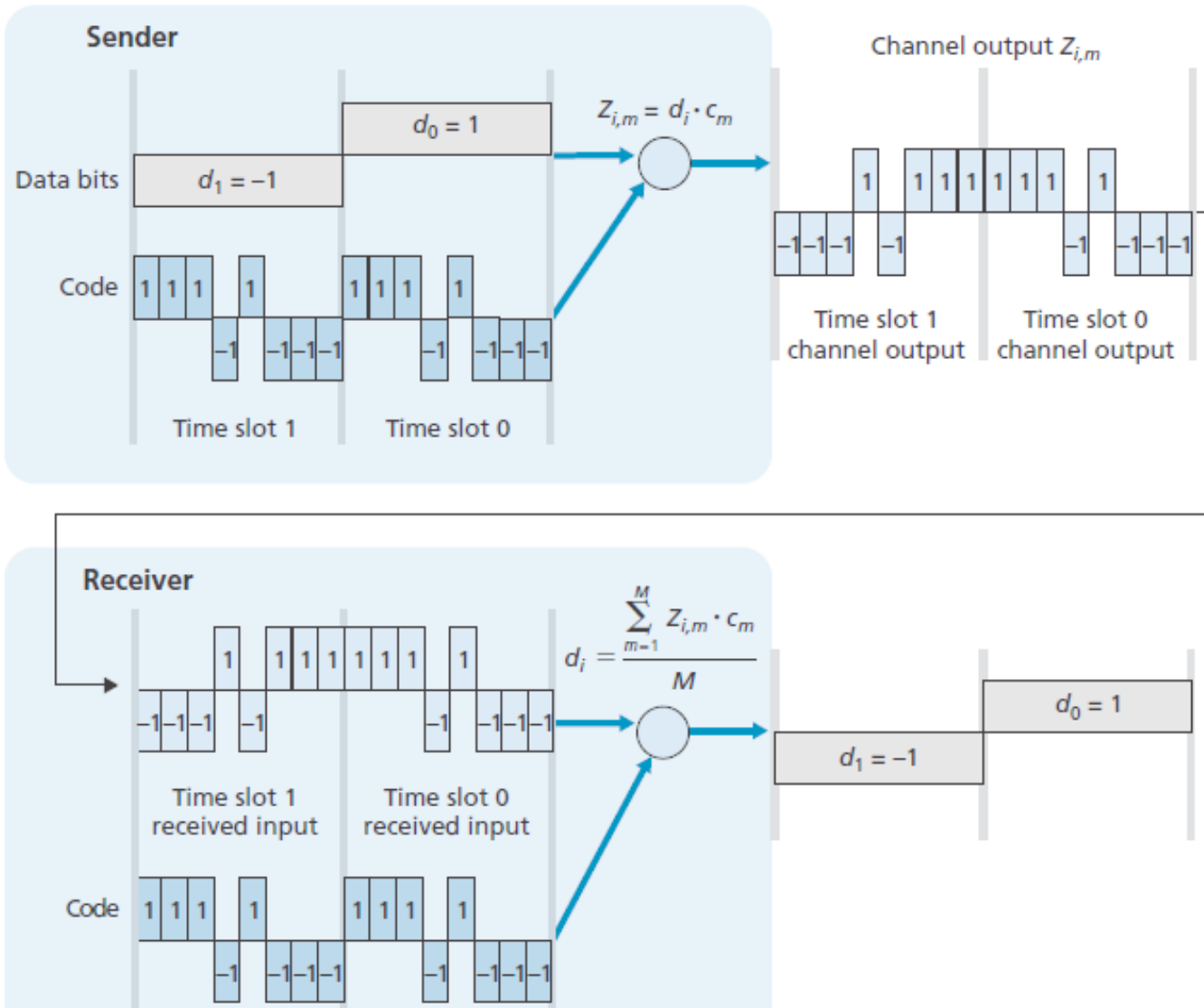
spread spectrum channels

DSSS (Direct Sequence Spread Spectrum)

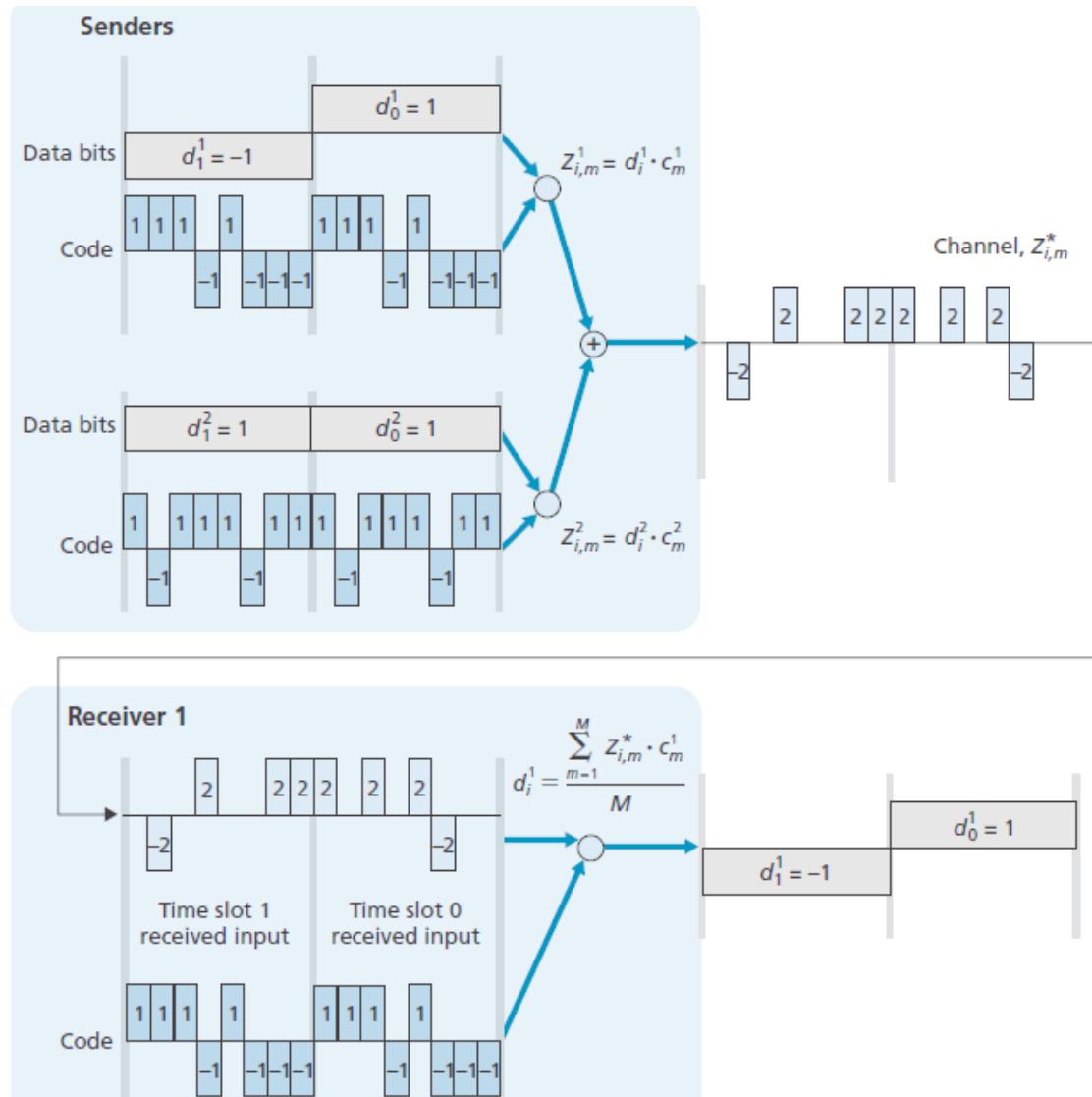
- XOR of the signal with pseudo-random number (chipping sequence)
 - many chips per bit (e.g., 128) result in higher bandwidth of the signal
- Advantages
 - reduces frequency selective fading
 - in cellular networks
 - base stations can use the same frequency range
 - several base stations can detect and recover the signal
 - soft handover
- Disadvantages
 - precise power control necessary



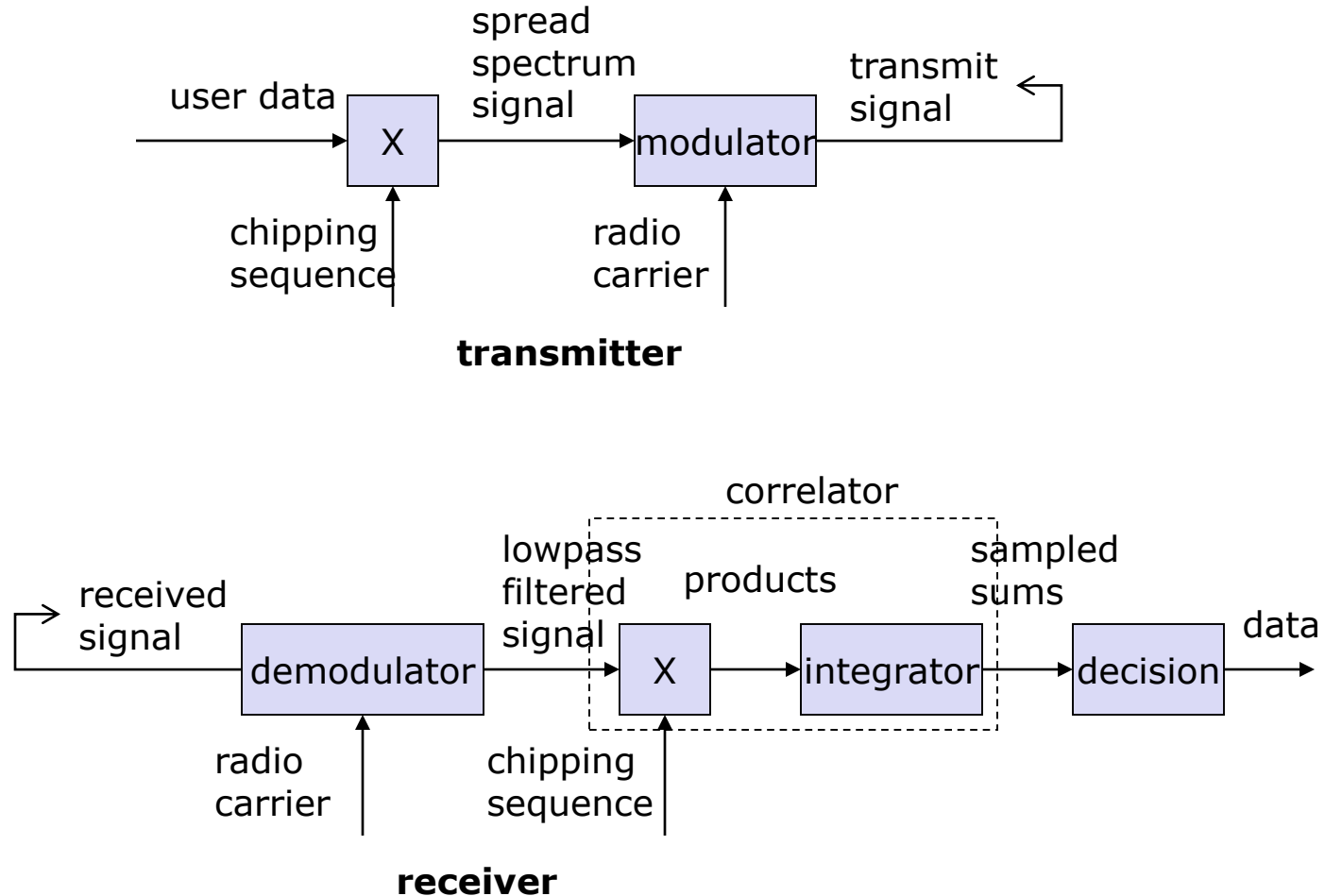
Example DSSS – Encoding and Decoding



Example DSSS – CDMA



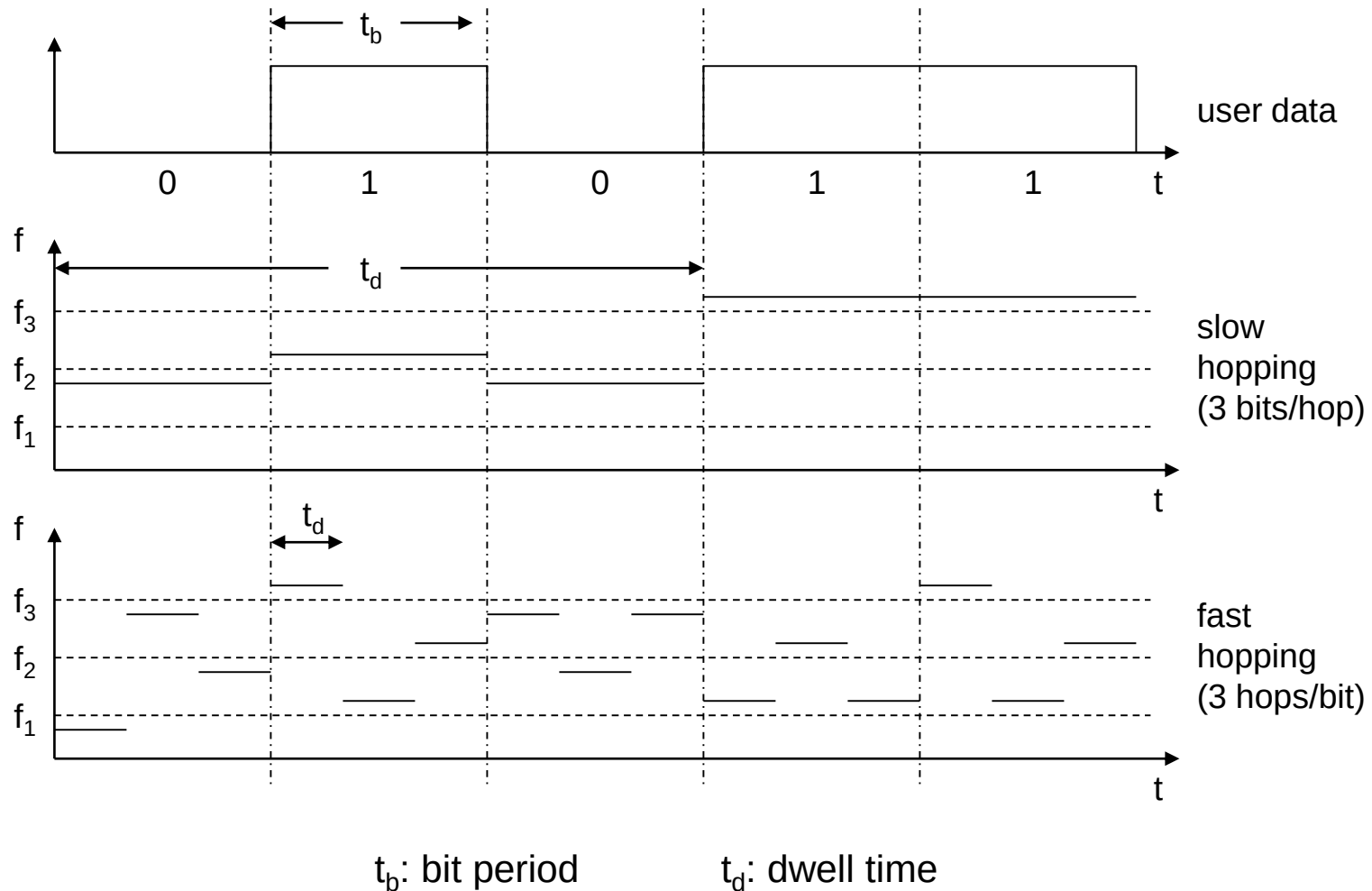
DSSS (Direct Sequence Spread Spectrum)



FHSS (Frequency Hopping Spread Spectrum)

- Discrete changes of carrier frequency
 - sequence of frequency changes determined via pseudo random number sequence
- Two versions
 - Fast Hopping:
several frequencies per user bit
 - Slow Hopping:
several user bits per frequency
- Advantages
 - frequency selective fading and interference limited to short period
 - simple implementation
 - uses only small portion of spectrum at any time
- Disadvantages
 - not as robust as DSSS
 - simpler to detect

FHSS (Frequency Hopping Spread Spectrum)



FHSS (Frequency Hopping Spread Spectrum)

